

E-Commerce Sales Analysis

Project Documentation

1. Project Overview

1.1. Project Statement

E-commerce businesses generate large amounts of transactional data, but raw sales data alone **does not provide enough insight** for effective decision-making. Management needs to understand sales performance, profitability, product performance, geographic distribution, and customer purchasing behavior.

This project aims to analyze e-commerce sales data and transform it into meaningful business insights through data cleaning, exploratory data analysis, dimensional modeling, and interactive Power BI dashboards.

The analysis focuses on **identifying high-performing products** and categories, detecting loss-making areas, understanding sales and profit trends, and supporting data-driven business decisions.

1.2. Project Objectives

The main objectives of the project are to:

- Analyze overall sales and profitability performance.
- Identify **the best- and worst-performing product categories** and sub-categories.
- Analyze sales and profit **trends over time**.
- Evaluate **geographic sales** and profitability performance.
- Understand customer purchasing **behavior**.
- Analyze **payment method performance**.
- Identify **loss-making product** segments and locations.
- Build an interactive Power BI dashboard for **business decision-making**.
- Transform the raw transactional data into **an optimized Star Schema**.

2. Dataset

2.1. Dataset Description

The dataset contains e-commerce transaction information covering **orders, products, customers, locations, dates, quantities, sales amounts, profits, and payment methods**.

The original transactional data was **transformed and organized into a dimensional model** to support efficient analysis and visualization in Power BI.

The final analytical model **separates transactional information** from descriptive attributes using a **Star Schema**.

2.2. Dataset Source

The dataset was obtained from an **e-commerce sales dataset** and was prepared and curated for the purpose of this analysis.

The data was **processed before analysis** to ensure consistency, remove duplicate records, correct data types, and prepare the tables for analytical modeling.

2.3. Key Variables

The main variables used in the analysis include:

- Order ID
- Order Date
- Customer ID
- Product information
- Category
- Sub-Category
- State
- City
- Quantity
- Sales Amount

- Profit
- Payment Mode

These variables allow the project to analyze the business **from multiple perspectives**, including time, product, geography, customers, and payment methods.

3. Data Dictionary

Variable	Description	Business Meaning
Order ID	Unique identifier for each order	Used to count and analyze orders
Order Date	Date on which the order was placed	Used for time-based analysis
Customer ID	Unique identifier for each customer	Used to analyze customer behavior
Category	High-level product classification	Used to compare product categories
Sub-Category	Detailed product classification	Used for detailed product analysis
Product	Product purchased by the customer	Used to analyze product performance
Quantity	Number of units purchased	Measures sales volume
Sales Amount	Monetary value of the transaction	Measures revenue generated
Profit	Profit generated from the transaction	Measures profitability
State	Customer/order state	Used for geographic analysis
City	Customer/order city	Enables detailed geographic analysis
Payment Mode	Method used to complete the payment	Used to analyze payment behavior
Month	Month extracted from the order date	Used for monthly analysis
Year	Year extracted from the order date	Used for yearly comparison

4. Data Cleaning & Preparation

The dataset was reviewed to ensure that it was suitable for analysis and Power BI modeling. The data did not contain significant missing values or incorrect data types that required correction. Therefore, no imputation or major data type conversion was necessary.

The main preparation steps focused on **structural organization and transformation** rather than correcting data quality issues.

4.1. Data Quality Validation

The dataset was checked for:

- Missing values
- Duplicate records
- Data type consistency
- Consistency of categorical values
- Validity of dates and numerical fields

The data was found to be generally consistent and suitable for analysis.

4.2. Data Transformation

The main transformations focused on preparing the dataset for the analytical model, including:

- Separating transactional data from descriptive attributes.
- Combining the Orders and Details data where required.
- Creating the **FactDetails** table.
- Creating dimension tables such as **DateDim**, **ProductDim**, **LocationDim**, **PaymentDim** and **CustomerDim**.
- Removing redundant descriptive columns from the fact table.
- Replacing repeated location information with **Location ID**.
- Replacing repeated payment Modes with **PaymentID**.

- Replacing repeated category & sub-category with **ProductID**.
- Replacing repeated customer names with **CustomerID**.
- Creating the Date Dimension and required date attributes such as Year, Month Number, Month Name, and Year-Month.
- Creating relationships between the fact and dimension tables.

4.3. Duplicate Handling

Duplicate records were reviewed during the preparation stage. Only records that were considered redundant based on the final modeling requirements were removed. No major duplicate-related data quality issue was identified.

5. Data Modeling - Star Schema

To improve data organization and analytical performance, the original transactional structure was transformed into a Star Schema.

The central fact table contains measurable business transactions, while dimension tables contain descriptive attributes.

5.1. Fact Table

FactDetails

Contains **transactional measures** such as:

- Order ID
- Order Date
- Customer ID
- Location ID
- Product ID
- Payment ID
- Amount
- Profit
- Quantity

5.2. Dimensions

LocationDim

- LocationID
- City
- State

ProductDim

- ProductID
- Category
- Sub-Category

DateDim

- Order Date
- Year
- Quarter
- MonthNo
- MonthName
- YearMonth

CustomerDim

- CustomerID
- CustomerName

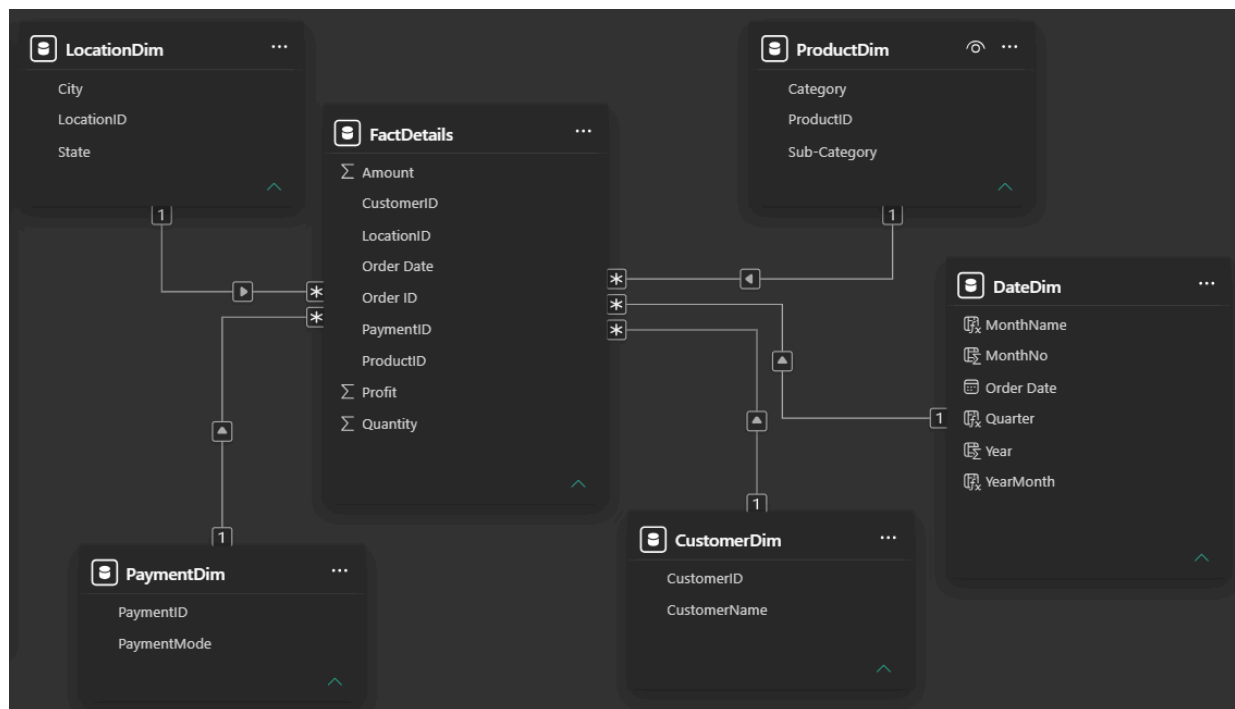
PaymentDim

- PaymentID
- PaymentMode

5.3. Relationships

Each dimension table is connected to factDetails through a **one-to-many relationship**, where the dimension represents the "one" side and the fact table represents the "many" side.

This structure allows the dashboard to analyze transactional measures across different dimensions such as time, product, location, customer, and payment method.



6. Measures & DAX

Several DAX measures were created to support the dashboard analysis.

Total Sales:

Total Sales =SUM(FactDetails[Amount])

Total Profit:

Total Profit =SUM(FactDetails[Profit])

Total Quantity:

Total Quantity =SUM(FactDetails[Quantity])

Total Orders:

Total Orders =DISTINCTCOUNT(FactDetails[Order ID])

Total Customers:

Total Customers =DISTINCTCOUNT(FactDetails[CustomerID])

Profit Margin:

Profit Margin =DIVIDE([Total Profit],[Total Sales],0)

Average Order Value:

Average Order Value =DIVIDE([Total Sales],[Total Orders],0)

Average Profit Per Order:

Average Profit per Order =DIVIDE([Total Profit],[Total Orders],0)

Repeat Customers:

Repeat Customers=

COUNTROWS(FILTER(VALUES(CustomerDim[CustomerID]),CALCULATE(DISTINCTCOUNT(FactDetails[Order ID])) > 1))

Repeat Customers Rate:

Repeat Customer Rate =DIVIDE([Repeat Customers],[Total Customers],0)

Loss-Making Sub-Categories:

Loss-Making Subcategories =

COUNTROWS(FILTER(VALUES(ProductDim[Sub-Category]),[Total Profit] < 0))

Loss-Making States:

Loss-Making States =

COUNTROWS(FILTER(VALUES(LocationDim[State]),[Total Profit] < 0))

Top City Sales:

Top City Sales = CALCULATE(MAX(LocationDim[City]),TOPN(1, ALL(LocationDim[City]), [Total Sales], DESC))

Top State Profit:

Top State Profit = CALCULATE(MAX(LocationDim[State]),TOPN(1, ALL(LocationDim[State]), [Total Profit], DESC))

Top State Sales:

Top State Sales = CALCULATE(MAX(LocationDim[State]),TOPN(1, ALL(LocationDim[State]), [Total Sales], DESC))

7. EDA

7.1. Data Overview

The dataset contains **1,500 records and 11 columns** after combining the Details and Orders datasets using a **left join** on Order ID.

The dataset includes information about orders, sales amount, profit, quantity, products, categories, payment methods, customers, dates, and geographic locations.

No missing values were found in the dataset, and no duplicate records were identified.

7.2. Data Type Validation

The data types of all variables were examined before performing the analysis.

The numerical variables **Amount, Profit, and Quantity** were already stored in suitable numerical formats. The categorical variables were also stored appropriately.

The **Order Date** column was initially stored as an object and was converted to **datetime format using day-first formatting** to support time-based analysis.

7.3. Unique Orders Analysis

Although the dataset contains **1,500 records**, only **500 unique Order IDs** were identified.

This difference is expected because a single order **can contain multiple products**, categories, or sub-categories. Therefore, the same Order ID can appear in multiple rows, with each row representing a different item or detail within the order.

For this reason, **Order ID** is treated as a unique identifier for orders rather than counting every row as a separate order.

7.4. Descriptive Statistics

Descriptive statistics were calculated for the main numerical variables: **Amount, Profit, and Quantity**.

The average amount was **291.85**, while the median was **122.00**. The average Profit was **24.64**, with a median of **8.00**. The average quantity was approximately **3.74 units**, while the median was **3 units**.

The relatively high standard deviation of Amount and Profit indicates considerable variation in transaction values and profitability. The values also show that the dataset contains both **profitable and loss-making transactions**.

```
df.describe()
```

	Amount	Profit	Quantity
count	1500.000000	1500.000000	1500.000000
mean	291.847333	24.64200	3.743333
std	461.924620	168.55881	2.184942
min	4.000000	-1981.00000	1.000000
25%	47.750000	-12.00000	2.000000
50%	122.000000	8.00000	3.000000
75%	326.250000	38.00000	5.000000
max	5729.000000	1864.00000	14.000000

7.5. Outlier Analysis

Boxplots were used to identify extreme observations in the numerical variables.

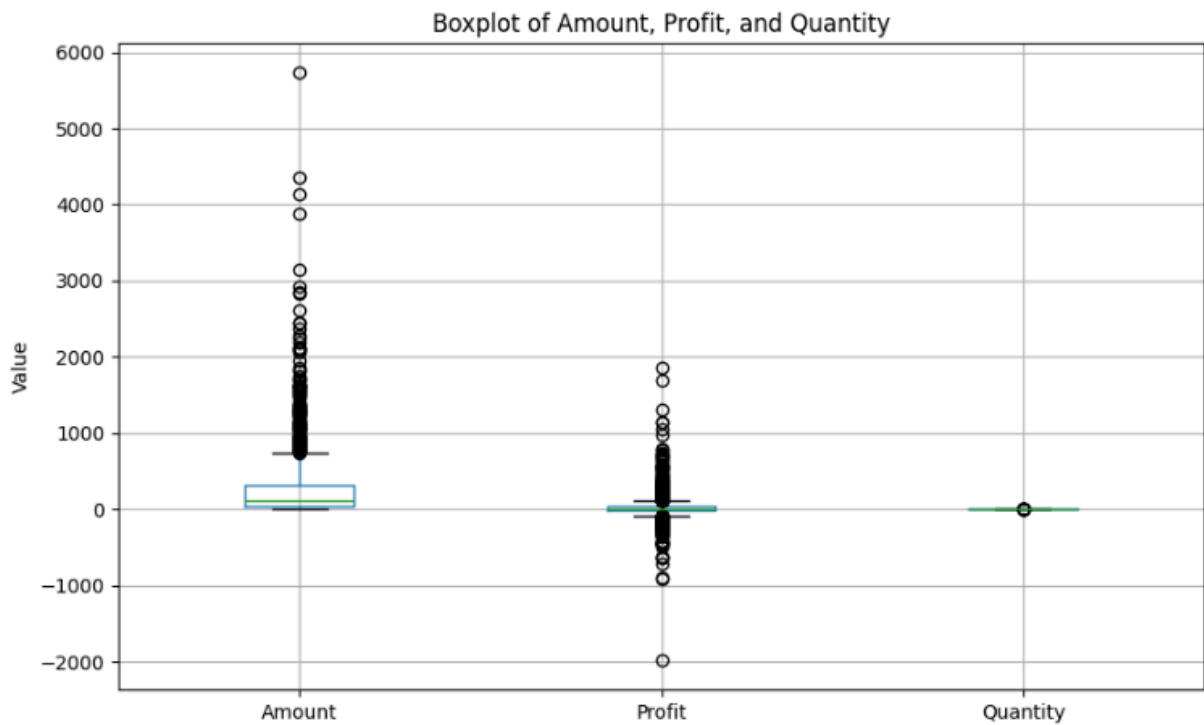
The analysis showed several extreme values, particularly in **Amount** and **Profit**. However, these observations were not removed because they may represent valid **high-value transactions or significant profitable and loss-making orders**.

Therefore, the detected extreme values were retained to preserve the **actual business behavior** represented in the dataset.

```
import matplotlib.pyplot as plt
plt.figure(figsize=(10, 6))

df[["Amount", "Profit", "Quantity"]].boxplot()

plt.title("Boxplot of Amount, Profit, and Quantity")
plt.ylabel("Value")
plt.xticks(rotation=0)
plt.show()
```



7.6. Correlation Analysis

A correlation matrix was created to examine the relationships between **Amount**, **Profit**, and **Quantity**.

The strongest relationship was between **Amount and Quantity**, with a correlation coefficient of **0.352**, indicating a weak-to-moderate positive relationship.

Amount and Profit had a positive correlation of **0.309**, while Profit and Quantity had a very weak correlation of **0.063**.

Overall, **no strong linear correlation was found** between the numerical variables. This suggests that **profitability is not determined by quantity alone** and may be **affected by other factors such as product category, sub-category, and location**.

<pre> numeric_cols = ["Amount", "Profit", "Quantity"] correlation = df[numeric_cols].corr() correlation </pre>			
	Amount	Profit	Quantity
Amount	1.000000	0.309242	0.352486
Profit	0.309242	1.000000	0.063096
Quantity	0.352486	0.063096	1.000000

7.7. Key EDA Findings

The EDA provided several important findings that guided the design of the Power BI dashboard:

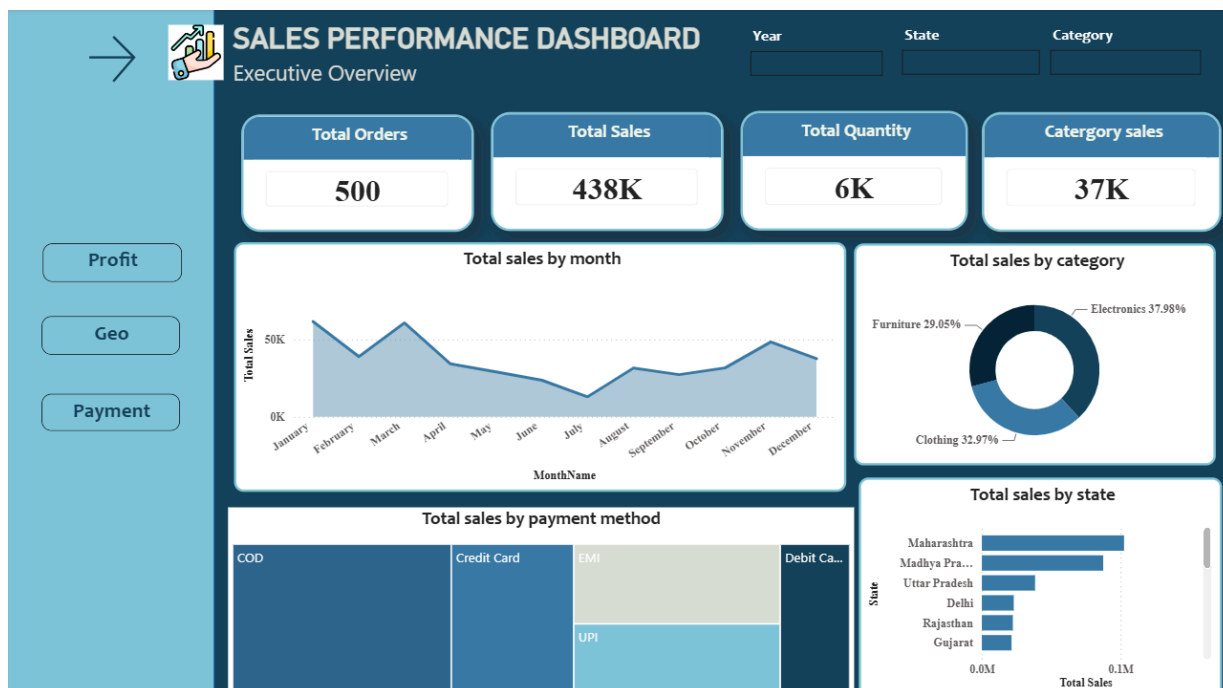
1. The dataset contains **1,500 transaction-level records representing 500 unique orders**.
2. The data contains **no missing values or duplicate records**.
3. The dataset includes both **profitable and loss-making transactions**.
4. Extreme values were detected but retained because they may represent valid business transactions.
5. **Clothing** has the largest number of transactions among the three categories.
6. **COD** is the most commonly used payment method.
7. **Madhya Pradesh and Maharashtra** have the highest transaction activity.
8. **Quantity has only a weak relationship with Profit**, indicating that selling more units does not necessarily result in proportionally higher profitability.

8. Power BI Dashboard

The final Power BI dashboard was designed to provide an interactive view of sales, profitability, product performance, geographic performance, and payment behavior.

The dashboard consists of **four analytical pages**, with each page focusing on a specific business perspective.

8.1. Page 1 - Sales Performance



Purpose

The Sales Performance page provides a high-level overview of the **business's sales activity**, allowing users to analyze sales, orders, quantity, profit, categories, states, and payment methods.

Main KPIs

- Total Sales
- Total Profit

- **Total Orders**
- **Total Quantity**

Main Visuals

- **Monthly Sales Trend**
- **Sales by State**
- **Sales by Category**
- **Sales by Payment Mode**

Business Questions

- What is the total sales revenue generated?
- What is the total number of orders?
- What is the total quantity of products sold?
- How do sales trend across the months of the year?
- Which states generate the highest sales?
- Which product categories contribute the most to sales?
- How are sales distributed across different payment modes?

Business Value

This page provides management with a quick overview of overall sales performance and allows them to identify major sales contributors across time, categories, locations, and payment methods.

8.2. Page 2 - Product & Profitability



Purpose

This page focuses on product performance and profitability, helping identify profitable and loss-making product segments.

KPIs

- Average Order Value
- Average Profit per Order
- Loss-Making Sub-Categories

Main Visuals

- Sales & Profit Trend by Month
- Sales & Profit Margin by Category
- Sales & Profit by Sub-Category
- Profit Distribution by Category
- Average Order Value vs. Profit Margin by Sub-Category

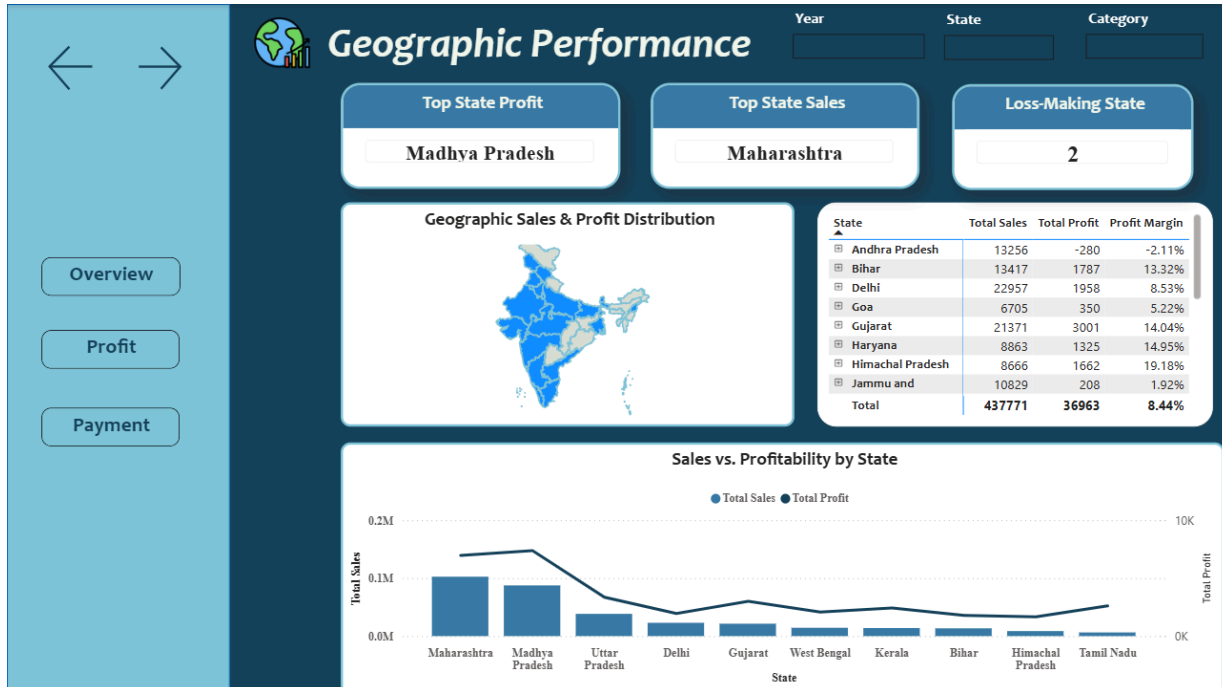
Business Questions

- How do sales and profit change throughout the year?
- Which product categories generate high sales while maintaining strong profit margins?
- Which sub-categories are the strongest contributors to sales and profit?
- How is total profit distributed across product categories?
- How many product sub-categories are generating losses?
- How does average order value relate to profitability across sub-categories?

Business Value

The page helps management distinguish between products that generate high sales and those that generate sustainable profitability. It also highlights loss-making sub-categories that may require further investigation.

8.3. Page 3 - Geographic Performance



Purpose

This page analyzes sales and profitability across different states and cities, helping identify strong and underperforming geographic markets.

KPIs

- **Top State Sales**
- **Top State Profit**
- **Loss-Making States**

Main Visuals

- **Geographic Sales & Profit Distribution Map**
- **Sales vs. Profitability by State**
- **State & City Performance Table**
- **State-to-City Drilldown**

The table allows users to examine **Total Sales, Total Profit, and Profit Margin** at both state and city levels.

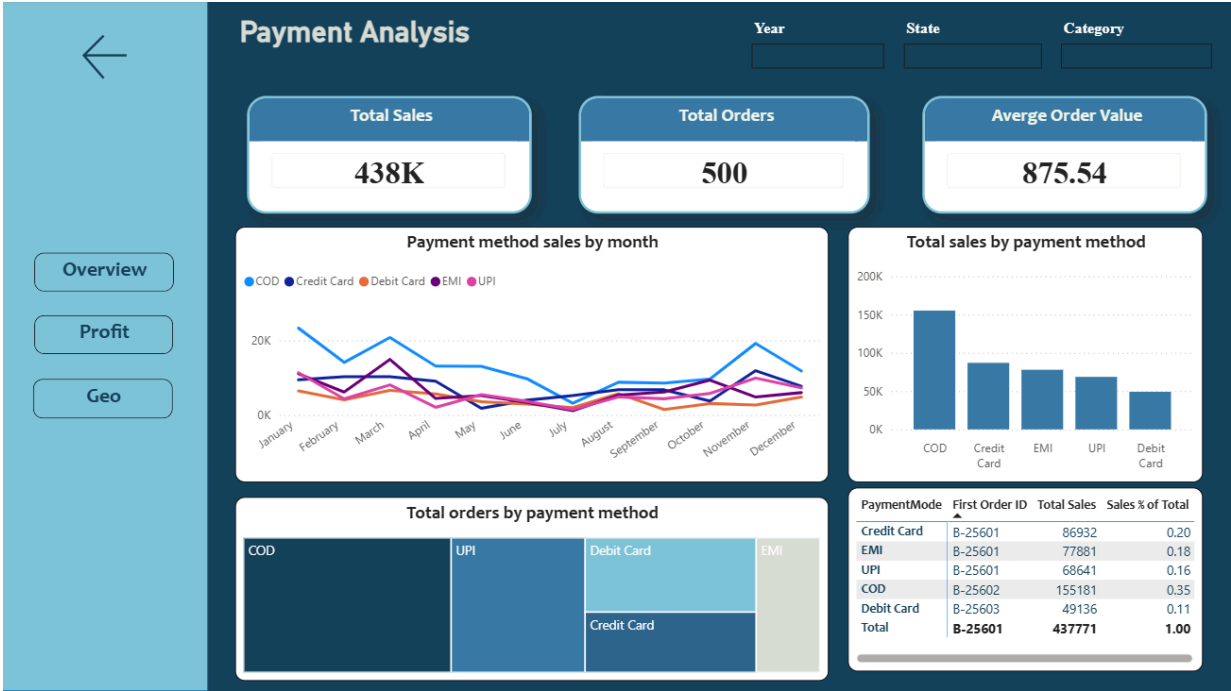
Business Questions

- Which states generate the highest sales?
- Which states generate the highest profit?
- Which states are loss-making?
- How are sales and profit distributed across different states?
- How does profitability compare with sales across states?
- Which states have high sales but relatively low profitability?

Business Value

Geographic analysis helps management identify strong markets, loss-making regions, and potential opportunities for targeted marketing and geographic expansion.

8.4. Page 4 - Payment Analysis



Purpose

This page focuses specifically on customer payment behavior and evaluates the contribution of different payment methods to sales and orders.

KPIs

- Total Sales
- Total Orders
- Average Order Value

Main Visuals

- Sales by Payment Mode
- Orders by Payment Mode
- Sales & Sales Contribution by Payment Mode
- Monthly Sales Trend by Payment Mode

Business Questions

- Which payment mode is used most frequently?
- Which payment mode generates the highest sales?
- What is the total sales value processed through each payment mode?
- What is the total number of orders per payment mode?
- What percentage of total sales comes from each payment mode?
- How do sales trends by payment mode change across the months?

Business Value

The page helps management understand customer payment preferences and identify the payment methods that contribute most to sales and order activity.

9. Dashboard Storytelling

The dashboard follows a **top-down analytical structure**, moving from overall business performance to more detailed analysis.

The user first starts with **Sales Performance** to understand the overall sales, orders, quantity, and distribution of sales across categories, states, and payment methods.

The analysis then moves to **Product & Profitability**, where users can investigate which categories and sub-categories are generating sales and profit and identify loss-making product segments.

Next, **Geographic Performance** provides a location-based view of business performance, allowing users to compare sales, profit, and profit margins across states and cities.

Finally, **Payment Analysis** focuses on payment behavior and shows how different payment modes contribute to sales and orders over time.

Interactive Filters

The dashboard provides interactive slicers that allow users to filter the analysis by relevant dimensions, including:

- **Year**
- **Category**
- **State**

Depending on the page, the available filters change to match the analysis being performed.

This interactivity allows users to move from an overall view to more focused analysis of specific categories, locations, years, or payment methods.

10. Key Findings

Based on the EDA and Power BI analysis, several important patterns were identified:

- The dataset contains **1,500 records representing 500 unique orders**, as multiple records can belong to the same order.
- **Clothing** represents the largest category by transaction volume.
- **COD** is the most frequently used payment method, followed by UPI.
- **Madhya Pradesh** has the highest transaction activity, while Maharashtra is also one of the strongest geographic contributors.
- Sales and profitability are not always directly aligned. A location or product segment can generate relatively high sales without achieving a similarly strong profit margin.
- Several sub-categories generate negative profit, highlighting areas that require further investigation.
- The correlation between **Quantity and Profit is very weak (0.063)**, indicating that selling more units does not necessarily result in higher profitability.
- Geographic performance varies across states, with some locations generating negative profitability despite generating sales.

11. Business Recommendations

Based on the findings from the EDA and Power BI dashboard, the following recommendations can support business decision-making:

1. Investigate Loss-Making Sub-Categories

Sub-categories generating negative profit should be analyzed to identify possible causes such as pricing, discounts, or product-related costs.

2. Focus on Profitable Product Segments

Products and categories that combine strong sales with healthy profit margins should receive greater attention in inventory and marketing decisions.

3. Optimize Geographic Strategy

High-performing states can be prioritized for expansion and marketing, while loss-making states should be investigated to understand the reasons behind their weak profitability.

4. Monitor Sales and Profit Trends

Monthly sales and profit trends should be monitored regularly to identify periods of strong or weak performance and support better planning.

5. Understand Payment Preferences

Since COD is the most frequently used payment method, the business should ensure a smooth COD experience while continuing to support other payment methods.

6. Focus on Profitability Rather Than Sales Alone

Management should avoid evaluating performance based only on sales volume. Profit margin and profitability should also be considered when making product and geographic decisions.

12. Limitations

The project has several limitations:

1. The analysis depends on the available transactional data.
2. The dataset contains limited customer and product attributes.
3. The analysis identifies relationships and patterns but does not establish causation.
4. External factors such as competitors, market conditions, and economic changes are not included.
5. The available historical data may not fully represent future business performance.
6. The dataset represents a limited period and may not capture long-term seasonal or market changes.

13. Conclusion

This project demonstrates how raw e-commerce transaction data can be transformed into a structured analytical solution that supports business decision-making.

Through data cleaning, exploratory data analysis, Star Schema modeling, DAX measures, and interactive Power BI dashboards, the project provides a comprehensive view of **sales performance, profitability, product performance, geographic performance, and payment behavior**.

The analysis highlights important business patterns, including differences in sales and profitability across categories, sub-categories, states, and payment methods. It also identifies loss-making product segments and geographic areas that may require further investigation.

The final dashboard enables management to move from a high-level view of overall performance to detailed analysis of **products, locations, time periods, and payment methods** through interactive filters and visualizations.

Overall, the analysis provides a data-driven foundation for improving **product strategy, pricing and profitability, geographic targeting, sales planning, and payment strategy**, helping the business make more informed decisions and identify opportunities for growth.